

0. Nomenclature

$\rho, \mu, \nu = \mu/\rho$	fluid density/viscosity/diffusivity	h grid size, dt step
$\mathbf{u}, p$	fluid velocity / pressure	$\mathbf{X}_s, \mathbf{V}_s$ solid position/velocity
$\mathbf{U}_l$	fluid velocity at marker	$\mathbf{F}_{\text{IBM}}, \Delta V_l$ force/marker volume
$\mathbf{F} = \nabla_X \mathbf{X}_s, J, \mathbf{P}$	deformation grad. / det / 1st P-K	$\mu_s, \lambda_s, \mu_s^{\text{visc}}$ solid moduli
$M_{\text{HRZ}}, V_{\text{node}}, K^{\text{visc}}$	lumped mass / node volume / visc. stiffness	ρ_s solid density

1. Incompressible NS with IBM force

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \mu \Delta \mathbf{u} + \mathbf{f}_{\text{IBM}}, \quad \nabla \cdot \mathbf{u} = 0$$

$\mathbf{f}_{\text{IBM}}$: immersed-boundary force spread to the fluid, non-zero only on marker nodes.

2. IB interpolation & spreading (Peskin 4-pt kernel)

$$\mathbf{U}_l = \sum_{\mathbf{x}} \mathbf{u}(\mathbf{x}) \delta_h(\mathbf{x} - \mathbf{X}_{s,l}) h^2 \quad (\text{fluid} \rightarrow \text{solid})$$

$$\mathbf{f}_{\text{IBM}}(\mathbf{x}) = \sum_l \mathbf{F}_{\text{IBM},l} \delta_h(\mathbf{x} - \mathbf{X}_{s,l}) \Delta V_l \quad (\text{solid} \rightarrow \text{fluid})$$

$$\delta_h(r) = \begin{cases} \frac{1}{4h} \left(1 + \cos \frac{\pi r}{2h} \right), & |r| \leq 2h \\ 0, & |r| > 2h \end{cases}$$

3. Time stepping (one step, partitioned explicit)

3.1 Predictor — AB2 convection + $\frac{3}{2} \frac{1}{2}$ semi-implicit diffusion: find $\mathbf{u}^* \in V$ s.t.

$$\begin{aligned} \int \mathbf{u}^* \cdot \mathbf{v} \, d\mathbf{x} + \frac{3}{2} \nu dt \int \nabla \mathbf{u}^* : \nabla \mathbf{v} \, d\mathbf{x} &= \int \mathbf{u}^n \cdot \mathbf{v} \, d\mathbf{x} - \frac{3}{2} dt \int (\mathbf{u}^n \cdot \nabla \mathbf{u}^n) \cdot \mathbf{v} \, d\mathbf{x} \\ &+ \frac{1}{2} dt \int (\mathbf{u}^{n-1} \cdot \nabla \mathbf{u}^{n-1}) \cdot \mathbf{v} \, d\mathbf{x} - \frac{1}{2} \nu dt \int \nabla \mathbf{u}^n : \nabla \mathbf{v} \, d\mathbf{x} + \frac{1}{2} dt \int \nabla p^n \cdot \mathbf{v} \, d\mathbf{x} \end{aligned}$$

3.2 Direct forcing (no-slip at markers, $\mathbf{U}_l \rightarrow \mathbf{V}_s$):

$$\mathbf{F}_{\text{IBM},l} = \frac{\mathbf{V}_{s,l} - \mathbf{U}_l}{dt} \Delta V_l, \quad \mathbf{u} = \mathbf{u}^* + dt \mathbf{f}_{\text{IBM}}$$

3.2b Multi-direct forcing (iterate until convergence):

$$\begin{aligned} \mathbf{u}^{(0)} &= \mathbf{u}^* \\ \mathbf{U}_l^{(k)} &= \sum_{\mathbf{x}} \mathbf{u}^{(k)} \delta_h(\mathbf{x} - \mathbf{X}_{s,l}) h^2, \quad \mathbf{F}_{\text{IBM},l}^{(k)} = \frac{\mathbf{V}_{s,l} - \mathbf{U}_l^{(k)}}{dt} \Delta V_l \\ \mathbf{u}^{(k+1)} &= \mathbf{u}^{(k)} + dt \sum_l \mathbf{F}_{\text{IBM},l}^{(k)} \delta_h(\mathbf{x} - \mathbf{X}_{s,l}) \Delta V_l \quad \text{until } \max_l \|\mathbf{U}_l^{(k)} - \mathbf{V}_{s,l}\| < \varepsilon \end{aligned}$$

3.3 Pressure Poisson:

$$\nabla^2 p^{n+1} = \frac{2}{3dt} \nabla \cdot \mathbf{u} \quad (\text{pin one corner dof})$$

3.4 L^2 projection (end-of-step divergence-free):

$$\mathbf{u}^{n+1} = \mathbf{u}^* - \frac{3}{2} dt \nabla p^{n+1}$$

4. Elastic solid (total Lagrangian, reference config)

4.1 Momentum:

$$\rho_s \frac{\partial^2 \mathbf{X}_s}{\partial t^2} = \nabla_X \cdot \mathbf{P}(\mathbf{F}) + \mathbf{f}^{f \rightarrow s}, \quad \mathbf{F} = \nabla_X \mathbf{X}_s$$

4.2 Constitutive (compressible neo-Hookean + Kelvin–Voigt viscosity):

$$\mathbf{P} = \mu_s (\mathbf{F} - \mathbf{F}^{-T}) + \lambda_s \ln J \mathbf{F}^{-T}, \quad J = \det \mathbf{F}$$

$$\boldsymbol{\sigma}^{\text{visc}} = 2 \mu_s^{\text{visc}} \text{sym}(\nabla_X \mathbf{V}_s), \quad \mu_s^{\text{visc}} = \mu$$

4.3 Internal force & HRZ lumped mass:

$$\mathbf{F}_{\text{int}} = \int \mathbf{P}(\mathbf{F}) : \nabla_X \delta \mathbf{v} \, dX, \quad M_{\text{HRZ},i} = M_{ii} \frac{M_{\text{tot}}}{\sum_j M_{jj}}, \quad M_{\text{tot}} = \rho_s \pi r^2$$

4.4 Solid advance (added-mass implicit; viscous term implicit in LHS):

$$[\text{diag}(M_{\text{HRZ}} + \rho_f V_{\text{node}}) + dt K^{\text{visc}}] \mathbf{V}_s^{n+1} = M_{\text{HRZ}} \mathbf{V}_s^n + \rho_f V_{\text{node}} \mathbf{U}_l - dt \mathbf{F}_{\text{int}}$$

$$K^{\text{visc}} = \int 2 \mu_s^{\text{visc}} \text{sym}(\nabla_X \delta \mathbf{u}) : \text{sym}(\nabla_X \delta \mathbf{v}) \, dX$$

$$\mathbf{X}_s^{n+1} = \mathbf{X}_s^n + dt \mathbf{V}_s^{n+1}$$

5. Rigid-disk variant (multi-direct-frocing, no elasticity)

$$\mathbf{V}_c = \langle \mathbf{U}_l \rangle_{\text{ring}}, \quad \boldsymbol{\omega} = \left\langle \frac{\mathbf{r} \times \mathbf{U}_l}{|\mathbf{r}|^2} \right\rangle_{\text{ring}}, \quad \mathbf{r} = \mathbf{x} - \mathbf{X}_c$$

$$\mathbf{U}^d(\mathbf{x}) = \mathbf{V}_c + \boldsymbol{\omega} \times \mathbf{r} \quad (\text{rigid constraint}), \quad \mathbf{X}_c^{n+1} = \mathbf{X}_c^n + dt \mathbf{V}_c, \quad \theta^{n+1} = \theta^n + dt \omega$$

6. Per-step algorithm

$$\mathbf{u}^{n-1}, \mathbf{u}^n, p^n \xrightarrow{3.1} \mathbf{u}^* \xrightarrow{3.2} \mathbf{F}_{\text{IBM}} = \frac{\mathbf{V}_s - \mathbf{U}_l}{dt} \Delta V_l \xrightarrow{3.3, 3.4} \mathbf{u}^{n+1}, p^{n+1}$$

$$(M_{\text{HRZ}} + \rho_f V_{\text{node}} + dt K^{\text{visc}}) \mathbf{V}_s^{n+1} = M_{\text{HRZ}} \mathbf{V}_s^n + \rho_f V_{\text{node}} \mathbf{U}_l - dt \mathbf{F}_{\text{int}}$$

7. Stability

$$\alpha = \frac{\rho_f V_{\text{node}}}{M_{\text{HRZ}} + \rho_f V_{\text{node}}} \quad (\text{added-mass ratio, } 0 < \alpha < 1)$$

$$dt < c \frac{h_s}{\sqrt{(\mu_s + \lambda_s)/\rho_s}} \quad (\text{elastic wave CFL}), \quad dt < c \frac{h^2}{\nu} \quad (\text{fluid diffusion})$$

viscous term implicit ($dt K^{\text{visc}}$ on LHS) $\Rightarrow$ unconditionally stable for the viscous part.